

ESWARVIKRAM E

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EDUCATION

Indian Institute of Technology Bombay <i>PhD in Water Resources Engineering</i>	Mumbai, India 2026 – Present
College of Engineering Guindy, Anna University <i>Master of Engineering in Irrigation Water Management</i>	Chennai, Tamil Nadu 2021 – 2023
Coimbatore Institute of Technology <i>Bachelor of Engineering in Civil Engineering</i>	Coimbatore, Tamil Nadu 2016 – 2020

PUBLICATIONS

Eswarvikram, E., Abdul Wahid, A., Arun Babu, E., Akshaya Prabha, A.V., and Gilbert Johnson, P. "A Statistical Evaluation of Sentinel-2 and Landsat-9 Sensors in Water Quality Retrieval in Mesotrophic and Hypereutrophic Reservoirs of India". Advances in Space Research, 2025.
DOI: 10.1016/j.asr.2025.03.042

EXPERIENCE

Indian Institute of Technology Bombay <i>Project Associate-1</i>	Feb 2026 – July 2026 Mumbai, India
Worked on LULC change and their correlation with Rainfall and Temperature patterns using Satellite Remote sensing datasets.	
National Institute of Technology Karnataka <i>Project Assistant</i>	Aug 2024 – Feb 2026 Surathkal, India
- Worked on multi-sensor integration for Soil quality retrieval, focusing on Soil Electrical Conductivity (EC) and Soil Organic Carbon (SOC) modeling. - Developed a data fusion models that combined in-situ Soil measurements, Remote sensing data, topographic variables derived from Digital Elevation Models, and Hydrometeorological parameters. - Mapped soil organic carbon using SENTINEL-1 SAR data and automated processing workflows in SNAP	
State Water Resources Management Agency <i>Water Resources Intern</i>	Nov 2022– Dec 2022 Chennai, TN
- Developed a comprehensive geodatabase system to track and manage data for over 100+ tanks in the Vilupuram District, resulting in a more efficient and organized water resources management process - Calculated Evapotranspiration rates for crops using CROPWAT 8.0 software, resulting in accurate water consumption estimates and improved irrigation management.	
AICTE - Jal Dharohar Sanrakshan Internship <i>Student Intern</i>	July 2022 – August 2022 Chennai, TN
- Collaborated with a team of students to conduct in-depth research and analysis on the current state of the water body, resulting in the identification of key areas for targeted restoration efforts. - Developed detailed maps utilizing ArcGIS to visualize water quality trends over time, allowing for more targeted decision-making and leading to a reduction in contamination levels.	

PROJECTS

Quantitative retrieval and spatiotemporal analysis of soil quality parameters using geospatial datasets and machine learning techniques in Western Plains and Ghat regions of Karnataka

- Worked on multi-sensor integration for Soil quality retrieval, focusing on Soil Electrical Conductivity (EC) and Soil Organic Carbon (SOC) modeling.
- Developed a data fusion models that combined in-situ Soil measurements, Remote sensing data, topographic variables derived from Digital Elevation Models, and Hydrometeorological parameters.

Evaluation of Water Quality Retrieval Equations Using Multispectral Remote Sensing Imageries in Bhavanisagar reservoir | *Python, Google Earth Engine, ArcGIS, laboratory Analysis*

- Monitored water quality of optical active parameters in Bhavanisagar Reservoir by comparing in-situ measurements with the results of Landsat-9 (OLI-2) and Sentinel-2 (MSI) data.
- Utilized Google Earth Engine for image processing and surface reflectance Extraction.
- Developed regression equations for Chl-a, EC, and TDS with high accuracy.
- Identified Sentinel-2's superior performance over Landsat-9 for enhanced spatial resolution.

A Comparative Analysis of Sentinel-2 and Landsat-9 Sensors for Water Quality Retrieval in Mesotrophic and Hypereutrophic Reservoirs in India. | *Python, Google Earth Engine, ML*

- Executed a focused study on BhavaniSagar and Krishnagiri reservoirs, extracting surface reflectance from LANDSAT-9 and SENTINEL-2 imagery using Google Earth Engine, correlating band reflectance values with Optical water quality parameters (Chlorophyll a, EC, TDS).

SKILLS

Technical: Remote Sensing, GIS, Hydrology, Irrigation, Water Quality, ML, Statistics

Software: ArcGIS, QGIS, SPSS, SNAP, Python

Developer Tools: Google Earth Engine, VS Code

Libraries: pandas, NumPy, Matplotlib, rasterio